

1. Source Code

counter.h

```
#ifndef COUNTER_H

#define COUNTER_H

void counter_init(int start);

int counter_next(void);

int counter_peek(void);

#endif
```

counter.c

```
#include "counter.h"

static int value = 0;

void counter_init(int start)
{
    int value = start;
}

int counter_next(void)
{
    value++;
    return value;
}

int counter_peek(void)
{
    return value;
}
```

main.c

```
#include <stdio.h>

#include "counter.h"

int main()
{
    counter_init(10);

    printf("%d\n", counter_next());

    printf("%d\n", counter_next());

    printf("%d\n", counter_peek());
}
```

```
    return 0;

}
```

2. Compilation Evidence

```
ztoptas@MSI:/mnt/c/Users/msi/OneDrive/Desktop/CSE101/prelabs/prelab11/try_again$ gcc -Wall -g counter.c
main.c -o program
```

counter.c: In function 'counter_init':

counter.c:7:9: warning: unused variable 'value' [-Wunused-variable]

```
7 |   int value = start; // BUG: shadows static variable
  |   ^~~~~~
```

3 & 4.

```
in$ gdb ./program
```

GNU gdb (Ubuntu 15.0.50.20240403-0ubuntu1) 15.0.50.20240403-git

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Type "show copying" and "show warranty" for details.

This GDB was configured as "x86_64-linux-gnu".

Type "show configuration" for configuration details.

For bug reporting instructions, please see:

<<https://www.gnu.org/software/gdb/bugs/>>.

Find the GDB manual and other documentation resources online at:

<<http://www.gnu.org/software/gdb/documentation/>>.

For help, type "help".

Type "apropos word" to search for commands related to "word"...

Reading symbols from ./program...

```
(gdb) break counter_init
```

Breakpoint 1 at 0x1154: file counter.c, line 7.

```
(gdb) break counter_next
```

Breakpoint 2 at 0x1165: file counter.c, line 12.

```
(gdb) run
```

Starting program: /mnt/c/Users/msi/OneDrive/Desktop/CSE101/prelabs/prelab11/try_again/program

This GDB supports auto-downloading debuginfo from the following URLs:

<<https://debuginfod.ubuntu.com>>

Enable debuginfod for this session? (y or [n]) y

Debuginfod has been enabled.

To make this setting permanent, add 'set debuginfod enabled on' to .gdbinit.

Downloading separate debug info for system-supplied DSO at 0x7ffff7fc3000

[Thread debugging using libthread_db enabled]

Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, counter_init (start=10) at counter.c:7

```
7      int value = start; // BUG: shadows static variable
```

(gdb) info args

start = 10

(gdb) info locals

value = 32767

(gdb) print value

\$1 = 32767

(gdb) print &value

\$2 = (int *) 0x7fffffdd8c

(gdb) info frame

Stack level 0, frame at 0x7fffffdda0:

rip = 0x55555555154 in counter_init (counter.c:7);

saved rip = 0x5555555519e

called by frame at 0x7fffffddb0

source language c.

Arglist at 0x7fffffdd90, args: start=10

Locals at 0x7fffffdd90, Previous frame's sp is 0x7fffffdda0

Saved registers:

rbp at 0x7fffffdd90, rip at 0x7fffffdd98

(gdb) backtrace

#0 counter_init (start=10) at counter.c:7

#1 0x00005555555519e in main () at main.c:5

(gdb) step

```
8      }
```

(gdb) info variables

All defined variables:

File ../csu/abi-note.c:

```
71:  static const struct {  
    Elf64_Nhdr nhdr;  
    char name[4];  
    int32_t desc[4];  
} __abi_tag;
```

File ../login/utmp_file.c:

```
37:  static int file_fd;  
39:  static off64_t file_offset;  
38:  static _Bool file_writable;  
42:  static struct utmp last_entry;
```

File ../nptl_db/db_info.c:

```
111:  const uint32_t _thread_db_const_thread_area;
```

File ../nptl_db/structs.def:

```
82:  const uint32_t _thread_db___nptl_initial_report_events[3];  
80:  const uint32_t _thread_db___nptl_nthreads[3];  
84:  const uint32_t _thread_db___pthread_keys[3];  
98:  const uint32_t _thread_db_dtv_dtv[3];  
116: const uint32_t _thread_db_dtv_slotinfo_list_slotinfo[3];  
95:  const uint32_t _thread_db_link_map_l_tls_modid[3];  
96:  const uint32_t _thread_db_link_map_l_tls_offset[3];  
66:  const uint32_t _thread_db_list_t_next[3];  
67:  const uint32_t _thread_db_list_t_prev[3];  
56:  const uint32_t _thread_db_pthread_cancelhandling[3];  
60:  const uint32_t _thread_db_pthread_eventbuf[3];  
61:  const uint32_t _thread_db_pthread_eventbuf_eventmask[3];  
62:  const uint32_t _thread_db_pthread_eventbuf_eventmask_event_bits[3];  
93:  const uint32_t _thread_db_pthread_key_data_level2_data[3];
```

```
52:  const uint32_t _thread_db_pthread_list[3];
63:  const uint32_t _thread_db_pthread_nextevent[3];
53:  const uint32_t _thread_db_pthread_report_events[3];
58:  const uint32_t _thread_db_pthread_schedparam_sched_priority[3];
57:  const uint32_t _thread_db_pthread_schedpolicy[3];
59:  const uint32_t _thread_db_pthread_specific[3];
55:  const uint32_t _thread_db_pthread_start_routine[3];
```

--Type <RET> for more, q to quit, c to continue without paging--q

Quit

(gdb) info variables counter

All variables matching regular expression "counter":

File ../sysdeps/posix/sprofil.c:

```
59:  static unsigned int overflow_counter;
```

File ./posix/register-atfork.c:

```
31:  static uint64_t fork_handler_counter;
```

Non-debugging symbols:

0x00007ffff7e04728 modcounter

(gdb) info variables counter.c

All variables matching regular expression "counter.c":

(gdb) info variables static

All variables matching regular expression "static":

(gdb) print 'counter.c'::value

\$3 = 0

(gdb) step

main () at main.c:7

```
7      printf("%d\n", counter_next());
```

(gdb) info locals

No locals.

(gdb) backtrace

#0 main () at main.c:7

(gdb) print 'counter.c'::value

\$4 = 0

(gdb) continue

Continuing.

Breakpoint 2, counter_next () at counter.c:12

12 value++;

(gdb) info locals

No locals.

(gdb) print 'counter.c':value

\$5 = 0

(gdb) continue

Continuing.

1

Breakpoint 2, counter_next () at counter.c:12

12 value++;

(gdb)

5.

Visible Variables at Each Breakpoint

At counter_init (Line 7):

- start (Argument) = 10
- value (Local) = 32767 (address: 0x7ffffffdd8c)
- value (Static) = 0

At counter_next (Line 12):

- value (Static) = 0

Variables that Disappear After Function Return

After counter_init returns:

- start (argument) - destroyed
- value (local) - destroyed

The static value remains unchanged at 0.

In counter_init, int value = start; creates a local variable that shadows the static value. The local variable is stored on the stack at 0x7ffffffdd8c and only exists during the function. When the function returns, this local value is destroyed without ever modifying the static value.

The static value stays at 0 because the assignment only affected the local value, not the static one. When `counter_next()` runs, it increments static value from 0 to 1 to 2, producing output 1, 2, 2 instead of the expected 11, 12, 12.